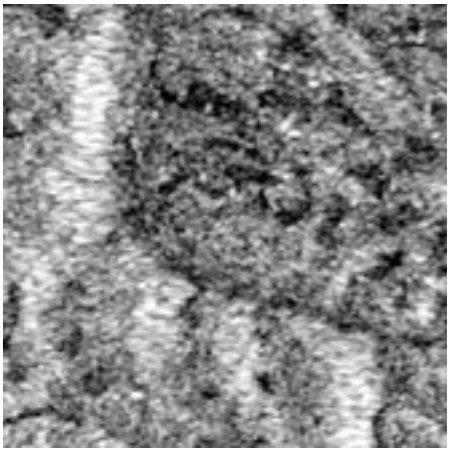
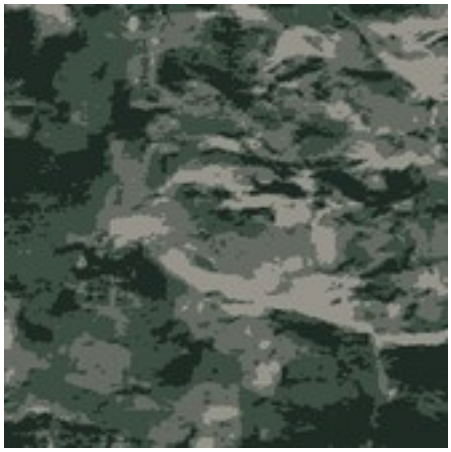
SAR Image Analysis Report



Original SAR

Colorized



K-Means Segmented

Metadata:

- Original File: original_1764348887.png
- Colorized File: colorized_1764348887.png
- Processed On: 2025-11-28 22:25:05 IST
- SAR Classifier Confidence: 0.997
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Vegetation (Green Tone) | Mean RGB: (101, 107, 98) | Area: 27.85%

Cluster 1: Likely Vegetation (Green Tone) | Mean RGB: (61, 78, 66) | Area: 36.19%

Cluster 2: Likely Urban / Built-up (Bright/Gray) | Mean RGB: (150, 148, 138) | Area: 11.94%

Cluster 3: Likely Vegetation (Green Tone) | Mean RGB: (31, 43, 37) | Area: 24.02%

Land Cover Summary:

- Likely Vegetation (Green Tone) → 27.85%
- Likely Vegetation (Green Tone) → 36.19%
- Likely Urban / Built-up (Bright/Gray) → 11.94%
- Likely Vegetation (Green Tone) → 24.02%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely vegetation (green tone) (88.06% combined).

Secondary presence: likely urban / built-up (bright/gray) (11.94%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is vegetation (36.29%). Strong vegetation signal suggests this area is suitable for agricultural monitoring — consider crop health and biomass monitoring. Secondary clusters show distributed variations; using time-series imagery would help confirm seasonal changes. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.997
Dominant Class Percentage: 36.19%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.